

Evidence-Bounded Post-Alert OT Incident Investigation with Electrical and Network Evidence

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Abstract—Process-aware intrusion detectors can identify suspicious deviations in operational technology (OT) systems, but an alarm does not establish what occurred, whether a command was executed, whether a physical operation completed, or whether a process change was caused by observed network activity. We study post-alert, event-conditioned OT investigation: an upstream source supplies an event scope and temporal anchor, and a language model generates evidence-linked claims from electrical/process and network observations. A deterministic verifier applies a frozen evidence-support policy to references, values, units, assets, timing, lineage, and claim-specific support requirements before publication. Using 32 pre-selected Sherlock v3 events and a fixed total retrieval budget, we compare electrical/process-only, network-only, and combined evidence. Combined evidence yields a Complete-Support Rate (CSR) of 0.9261, versus 0.7500 for electrical/process-only and 0.7406 for network-only, while exact gold-fact reconstruction remains 0.0000 in every view. Citation-required generation raises CSR from 0.8723 to 0.9261. Deterministic publication filtering raises verifier-aligned Citation Precision and CSR to 1.0000, while Q1–Q6 Coverage falls from 0.7760 to 0.5278. These results do not establish end-to-end detection or truth verification; they quantify a support–coverage trade-off under evidence integration and selective publication.

Index Terms—operational technology, industrial control systems, incident investigation, evidence grounding, large language models, deterministic verification

I. Introduction

Process-aware intrusion detection has advanced in identifying suspicious deviations in industrial systems, but an alarm is not an incident explanation. Sherlock evaluates industrial intrusion-detection systems using detected attacks, false alarms, and average time to detection, and shows that benign switching and maintenance can complicate detection in power-grid environments [1]. Such metrics establish whether and when a detector raises an alarm; they do not establish what occurred within an event, which observations support a conclusion, or which interpretations remain unjustified. Sherlock identifies actionability as an open challenge: systems should help operators understand and localize anomalous activity, not merely signal its presence.

This gap matters in OT because network and electrical/process observations have different semantics. A

network trace may show a command-type request, acknowledgement, or communication gap, while telemetry may show a reported value or state change. The electrical/process and network views are complementary representations, not independent sensors. Observing a command does not establish execution; an acknowledgement does not establish physical completion; and a process change after network activity does not establish causation, malicious intent, compromise, or attribution.

We study the stage after an alert has been raised. Our task is post-alert, event-conditioned OT investigation: an upstream detector, operational alert, or analyst supplies an event scope and temporal anchor t_0 , and the system determines which claims are supported by the electrical/process and network evidence visible within that scope. We do not evaluate continuous detection, unknown-onset discovery, or attack-versus-benign classification, and we do not independently establish causal truth, hidden physical state, intent, attack success, or attribution.

The pipeline normalizes heterogeneous observations into traceable records, retrieves evidence under a fixed view and budget, generates structured claims linked to evidence identifiers, and applies a deterministic verifier. The verifier applies a frozen evidence-support policy to visibility, references, numerical relationships, units, assets, timing, lineage, and claim-specific support requirements before publication. It assigns supported, qualified, or insufficient dispositions. It is not truth verification or causal inference and does not prove attack success, intent, or attribution.

We evaluate 32 preselected Sherlock v3 power-grid events with three planned repetitions per generative condition. Under a fixed total retrieval budget, we compare electrical/process-only, network-only, and combined views; compare citation-optional with citation-required generation under combined evidence; and replay exact raw combined-evidence outputs through the verifier without additional generation. Exact Evidence Completeness and View-Conditional Completeness are zero for every evidence view, so the study does not demonstrate exact reconstruction of frozen gold facts. Nevertheless, combined evidence yields higher CSR (0.9261) than electrical/process-only (0.7500) or network-only (0.7406),

while network-only has the highest Retrieval Recall. Requiring citations raises CSR from 0.8723 to 0.9261. Deterministic publication filtering raises verifier-aligned Citation Precision and CSR to 1.0000 while reducing Q1–Q6 Coverage from 0.7760 to 0.5278; V1 USCR is undefined because no security-sensitive substantive assertions are published. These findings characterize selective publication under a frozen evidence-support policy, not independent proof of factual correctness.

This work contributes (1) an evidence-bounded formulation of post-alert OT investigation with explicit semantic boundaries, (2) a controlled measurement of cross-view effects on retrieval, coverage, claim support, and exact reconstruction, and (3) a quantitative characterization of support–coverage trade-offs under citation-required generation and deterministic publication filtering. We address three research questions: RQ1, how do E, N, and combined ENviews affect reconstruction, retrieval, coverage, and support? RQ2, what changes when substantive claims must cite evidence? RQ3, what changes when raw ENoutput is exactly replayed through deterministic verification?

II. Background and Related Work

A. OT/ICS Intrusion Detection and Process-Aware Security

Process-aware monitoring uses knowledge of physical dynamics, invariants, or specifications to expose behavior that network features alone may not characterize; surveys review this research direction and specification-based systems encode safety constraints as security requirements [2], [3]. Cross-domain alert correlation likewise combines cyber-domain and physical-process detector outputs to construct richer scenarios [4]. Sherlock is particularly relevant because it benchmarks five process-aware IIDSs in power-grid environments using detected attacks, false alarms, and average time to detection [1]. Its results show that benign switching and maintenance complicate detection, and the authors identify actionability as a distinct challenge: helping operators understand and localize anomalous activity. Our study does not modify or outperform those detectors. Instead, it operationalizes a downstream aspect of actionability as evidence-bounded post-alert investigation: given a scoped event, which claims are supported by the electrical/process and network evidence visible to the investigator? Attack-level measures such as detected attacks and false alarms are not directly comparable with claim-conditioned measures such as CSR.

B. Incident Investigation and Explainable Detection

ICS response frameworks address uncertainty, coordination, and response planning [5], [6]. Explainable IDS work evaluates explanations of model decisions, including black-box explanation properties for network intrusion detection [7]. Such explanations differ from evidence-supported incident claims: feature attribution can explain

a classifier output without proving a physical action, command execution, or causal link. Our unit of analysis is the claim and the records required to support its wording.

C. LLMs for Cybersecurity Investigation

Surveys document LLM use across cybersecurity tasks [8]. Evidence-linked narratives are a related direction for security operations, but citation presence alone cannot ensure that cited records entail the full assertion. Hallucination research documents the broader risk of generated statements that are unsupported by the source context [9]. We pair synthesis with a deterministic verifier applying a finite frozen evidence-support policy, and evaluate support together with the useful content retained.

III. Problem Formulation and System Design

A. Event-Conditioned Investigation and Evidence Views

For event e , an external source supplies scope W_e and anchor t_0 . The visible bundle is $B_e^v = E_e^v \cup N_e^v \cup M_e^v$, where M contains sanitized static mappings and $v \in \{E, N, EN\}$. Ground truth G is evaluator-only and excluded from B_e^v . The investigator returns observations, temporal and asset relationships, bounded interpretations, and explicit unknowns. This is a scoped reconstruction task, not continuous detection.

B. Claim Semantics and Evidence Representation

Evidence records retain stable identifiers, timestamps, assets, units, source fields, and transformation lineage. Claim semantics constrain wording to the observation: a network record may support that a command-type request was observed; approved address metadata may support that it targeted a control point mapped to asset x . Neither proves asset execution. A response supports that an acknowledgement was observed, not physical completion. Electrical/process records support reported values and changes, not hidden physical truth or network causation. Timestamp ordering supports temporal association, not causality. Claims of intent, successful compromise, and attribution require support beyond these observations.

Table I gives representative boundaries. The LLM produces structured claims and evidence identifiers from retrieved records; it does not receive evaluator labels. A fixed view and evidence budget constrain retrieval. Citation-required output is compared with citation-optional output under otherwise fixed conditions.

C. Evidence-Grounded LLM Investigation

For each event and view, retrieval selects records within the externally supplied scope and the frozen input limits. The LLM structures observable facts, relationships, qualified interpretations, and unresolved questions. Substantive claims may cite evidence IDs; an ID must have been visible at generation time. Gold facts, attack labels, scenario descriptions, and hidden simulator state are inaccessible.

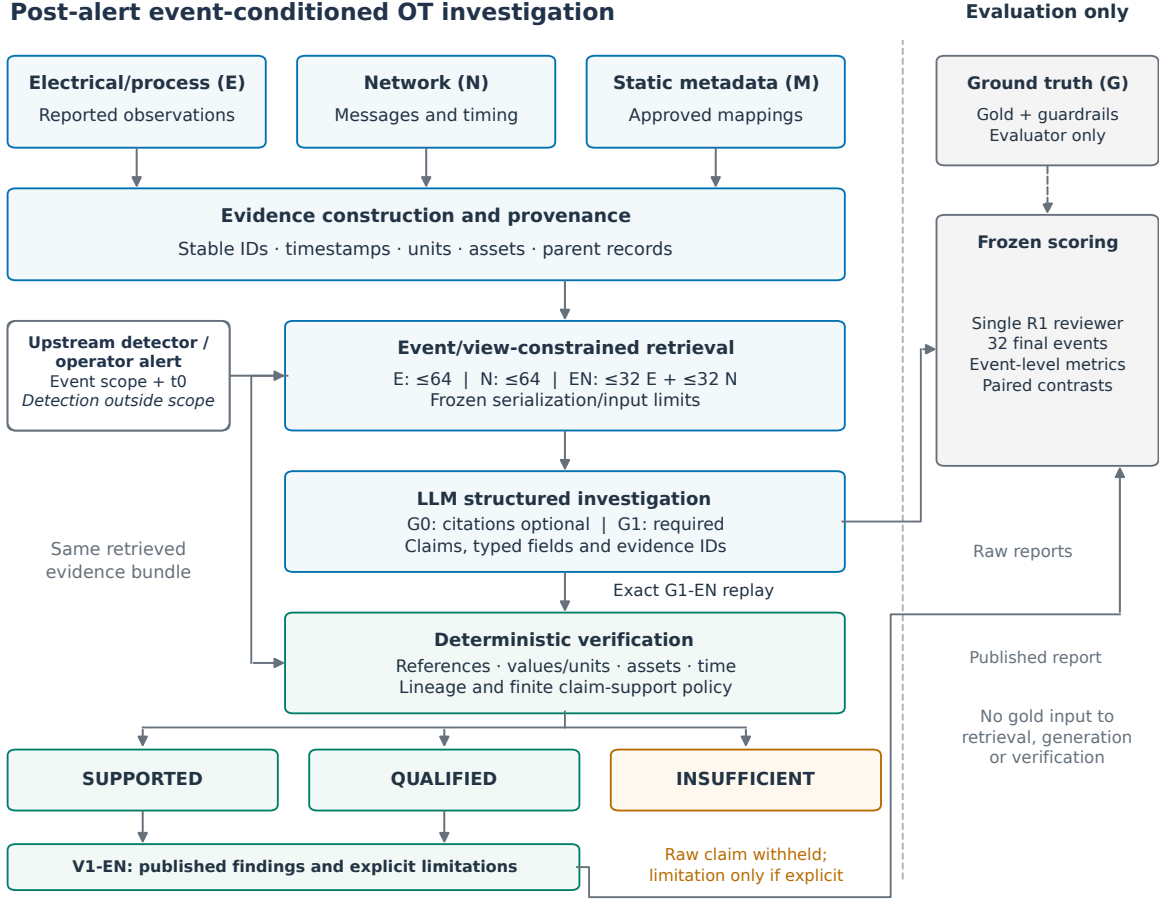


Fig. 1. Post-alert event-conditioned investigation, visibility boundary, deterministic verification, and evaluation flow.

TABLE I
Representative claim-support boundaries

Claim type	Evidence sufficient for bounded wording	Unsupported promotion
Network command	Command-type request observed; mapping permits “addressed to a control point mapped to x ”	Command executed; asset physically acted
Acknowledgement	Protocol response observed	Physical operation completed
Reported value/change	Electrical/process observation(s), compatible units, supported arithmetic	Hidden physical truth; network activity caused change
Asset/time relation	Approved mapping and timestamps establish relation/order	Compromise; temporal association implies causation
Security conclusion	Only a claim-specific policy with sufficient visible support	Intent, successful compromise, or attribution from observation alone

D. Deterministic Verification and Evidence Sufficiency

The deterministic verifier applies the frozen evidence-support policy to visible references, numerical values and units, asset mappings, time ordering, transformation lineage, and predefined claim-specific support requirements. It assigns supported, qualified, or insufficient dispositions. It is not truth verification or causal inference and does not prove attack success, intent, or attribution. A zero denominator for a claim-conditioned metric is NA, not zero; absence of support is not evidence of a benign event.

IV. Experimental Methodology

A. Dataset and Event Set

The study uses Sherlock v3, a Wattson power-grid simulation dataset with IEC 60870-5-104 traffic and electrical/process observations [1], [10]. Sixteen events were reserved for development and 32 preselected events for final evaluation. The final set is scored against reviewed event facts and unresolved-conclusion opportunities; evaluator-only material is separated from model input.

B. Experimental Conditions

Table II summarizes the cells. All generative cells use one fixed model snapshot (gpt-4.1-2025-04-14), frozen prompt/schema and decoding settings, and three planned repetitions per event. RQ1 holds the total retrieval budget at 64 records: 64 E, 64 N, or a 32 E + 32 N allocation for EN, with serialized-input limits also fixed. G1-E has three provider-incomplete repetitions (93 valid deliveries of 96); these are repetition-level NA and all 32 events remain defined. V1-EN is exact replay of raw G1-EN: no additional LLM generation.

TABLE II
Experimental condition summary

Cell	View	Role / policy
D0	EN	Deterministic reference; descriptive only
G0	EN	LLM; citations optional (RQ2)
G1-E	E	LLM; citations required (RQ1)
G1-N	N	LLM; citations required (RQ1)
G1-EN	EN	LLM; citations required (RQ1-RQ3)
V1-EN	EN	Exact G1-EN replay; verifier (RQ3)

C. Human Review and Metrics

A single R1 reviewer completed 477 corrected human-review items; no dual-review or inter-annotator-agreement claim is made. Metrics are first computed per repetition, averaged within event, then macro-averaged across events. We report exact Evidence Completeness, Retrieval Recall, Q1-Q6 Coverage, Citation Precision, CSR (complete cited support among supported substantive claims), Unsupported Security-Claim Rate (USCR), and withholding metrics. D0 is descriptive and excluded from primary RQ comparisons. For claim-conditioned metrics, a zero denominator is NA.

D. Statistical Analysis

Contrasts are paired at the event level. Confidence intervals are 95% percentile intervals from 2,000 scenario-stratified bootstrap resamples, seed 20270922. All final conditions have 32 evaluation events; generative conditions have three planned repetitions per event. We preserve the fixed matching criterion and do not treat claims or repetitions as independent samples.

V. Results

Exact reconstruction fails under the frozen matching criterion: Evidence Completeness and View-Conditional Completeness are 0.0000 for E, N, and EN. Other metrics describe stages of post-alert investigation, not successful incident reconstruction. The RQ1 funnel localizes this gap: the event-level means are 14 full gold facts, 14 supportable facts under EN, and 3.125 facts with complete support retrieved; reconstruction remains zero. This separates evidence availability, retrieval, support, and exact matching. EN has higher CSR (0.9261) than E (0.7500) and N (0.7406), and improves Q1-Q6 Coverage relative to N;

its coverage difference from E includes zero. Network-only has the highest Retrieval Recall, so EN does not dominate every stage. Table III summarizes outcomes and paired contrasts.

A. RQ1—Electrical and Network Evidence

The paired contrasts confirm this separation. EN improves CSR over both single views and Q1-Q6 Coverage relative to N, whereas its coverage difference from E includes zero. Network-only retains the highest Retrieval Recall.

B. RQ2—Evidence-Grounded Generation

Requiring citations primarily improves completeness of claim support: CSR increases from 0.8723 to 0.9261 (paired $\Delta = +0.0539$, 95% CI [0.0287, 0.0794]). Paired differences in Citation Precision and Q1-Q6 Coverage include zero. USCR is 0.0000 where defined in both conditions, but sparse denominators (13 G0 and 15 G1-EN defined events; seven jointly defined) do not establish that citation-required generation reduces unsupported security conclusions. G0 is citation-optional, not citation-free.

C. RQ3—Deterministic Verification

Deterministic publication filtering on exact replay raises verifier-aligned Citation Precision from 0.9949 to 1.0000 and CSR from 0.9261 to 1.0000, while Q1-Q6 Coverage falls from 0.7760 to 0.5278 ($\Delta = -0.2483$, 95% CI [-0.2934, -0.1979]). Required Withholding Recall is 0.3344; Withholding Precision is 1.0000 on defined events. V1 publishes no security-sensitive substantive assertions, so its USCR denominator is zero on all 32 events and the metric is NA, not zero. This is selective publication under the frozen evidence-support policy, not truth verification. Perfect verifier-aligned metrics accompany lower coverage, and withholding unsupported content does not ensure every required limitation is communicated.

TABLE III
Primary results (event-level macro means; paired contrasts with 95% bootstrap CIs)

RQ	Metric	Condition means	Primary paired contrast / denominator note
1	Evidence Completeness	E 0.0000; N 0.0000; EN 0.0000	EN-E 0.0000 [0.0000, 0.0000]; EN-N 0.0000 [0.0000, 0.0000]
1	View-Conditional Completeness	All 0.0000	No view reconstructs a frozen gold fact
1	Retrieval Recall	E 0.1111; N 0.3500; EN 0.2188	EN-E +0.1076 [0.0920, 0.1207]; EN-N -0.1313 [-0.1938, -0.0714]
1	Q1-Q6 Coverage	E 0.7691; N 0.5972; EN 0.7760	EN-E +0.0069 [-0.0686, 0.0747]; EN-N +0.1788 [0.0885, 0.2674]
1	Citation Precision	E 0.9901; N 0.9589; EN 0.9949	EN-E +0.0048 [-0.0048, 0.0192]; EN-N +0.0360 [0.0186, 0.0565]
1	Complete-Support Rate	E 0.7500; N 0.7406; EN 0.9261	EN-E +0.1761 [0.1449, 0.2030]; EN-N +0.1856 [0.1480, 0.2208]
2	Citation Precision; CSR; coverage	G0 0.9907; 0.8723; 0.7465. G1-EN 0.9949; 0.9261; 0.7760	CSR: +0.0539 [0.0287, 0.0794]. Citation precision: +0.0042 [-0.0027, 0.0121]; coverage: +0.0295 [-0.0313, 0.0903]
2	USCR	G0 0.0000; G1-EN 0.0000 when defined	Sparse: G0 13 defined/19 NA events; G1-EN 15/17; paired denominator 7. No decrease established
3	Citation Precision; CSR	G1-EN 0.9949; 0.9261. V1-EN 1.0000; 1.0000	V1-G1: +0.0051 [0.0015, 0.0092]; +0.0739 [0.0504, 0.0970]
3	Q1-Q6 Coverage	G1-EN 0.7760; V1-EN 0.5278	-0.2483 [-0.2934, -0.1979]
3	Withholding; USCR	V1 recall 0.3344; precision 1.0000 on defined events	Precision defined on 27 events (5 NA); raw USCR defined on 15 events, V1 USCR = NA on all 32

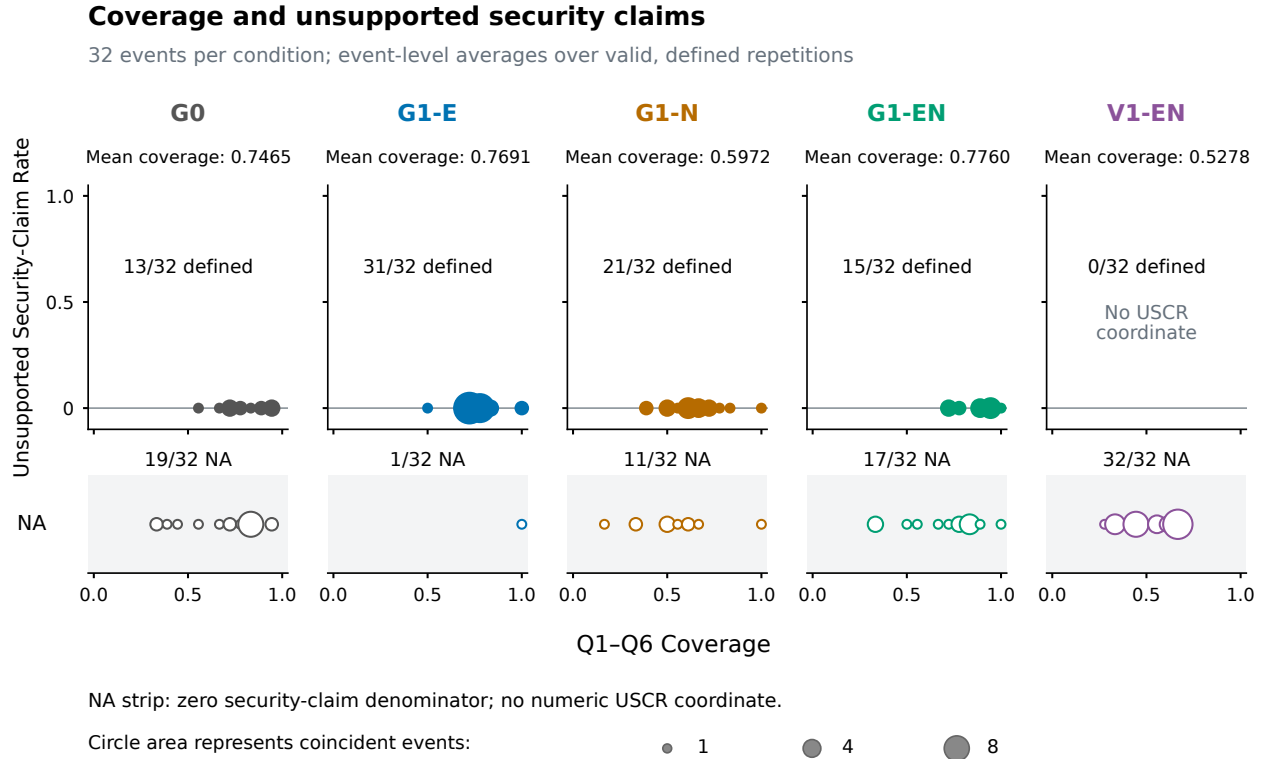


Fig. 2. Coverage and USCR by condition. For V1-EN, all 32 security-sensitive assertion denominators are zero; its USCR coordinate is NA, not zero. The NA strips in the source figure retain undefined positions.

VI. Discussion and Limitations

Evidence-support enforcement creates a measurable support-coverage trade-off in post-alert investigation. Combined electrical/process and network evidence improves support completeness for produced claims, yet neither evidence integration nor deterministic publication filtering solves exact event reconstruction. The zero-

reconstruction result bounds the claim: the pipeline can produce better-supported partial investigative claims without establishing complete incident reconstruction. V1's Citation Precision and CSR of 1.0000 reflect selective publication under the frozen evidence-support policy used to define admissible claims. They show policy-aligned consistency of the retained set, not independent factual

correctness, causal truth, or attribution. The coverage decline from 0.7760 to 0.5278 is essential: perfect support metrics are obtained in part by publishing less.

In the broader Sherlock setting, this work is complementary to process-aware intrusion detection, not a replacement. Sherlock shows that power-grid detectors must contend with maintenance, changing operating states, benign anomalies, and false alarms. Our evaluation starts after an event scope has been supplied and asks what can be stated from evidence inside that scope without promoting observations into unsupported execution, causation, or security conclusions. Connecting detector-generated alarm windows, including true-positive attacks and maintenance-related false alarms, to this downstream pipeline is an important next step toward end-to-end evaluation.

These results are limited to 32 preselected events from one power-grid co-simulation dataset, one fixed model snapshot, and one R1 reviewer. The single-reviewer design provides no inter-annotator-agreement estimate. Event-conditioned scope is supplied externally, so results do not measure continuous detection, unknown-onset discovery, or incident selection. The zero exact completeness limits claims about reconstruction even though coverage and support are measurable. The verifier covers finite evidence rules and cannot adjudicate real-world physical truth, causal effects, intent, successful compromise, or attribution. The lack of a defined V1 USCR prevents pre/post USCR comparison; other metrics also have condition-specific sparse or undefined denominators. Electrical/process values are reported observations rather than guaranteed physical truth, and E and N may share packet ancestry. Exact replay and frozen inputs strengthen within-study comparisons, but model behavior, structured output, retrieval, and the single-reviewer policy constrain generalization. Three provider-incomplete G1-E repetitions remain repetition-level NA. No analyst study was conducted; workload reduction or autonomous SOC readiness is not established. The experiment does not compare alternative verifier designs, so retaining more qualified content remains future work.

VII. Conclusion

We studied OT investigation after an upstream alert, asking which claims are supported by the electrical/process and network evidence visible within an externally supplied event scope. Combined evidence improves support completeness for produced claims, but exact gold-fact reconstruction remains zero in every evidence view. Citation-required generation increases CSR from 0.8723 to 0.9261. Deterministic publication filtering raises verifier-aligned Citation Precision and CSR to 1.0000 while reducing Q1–Q6 Coverage from 0.7760 to 0.5278. The central result is a measurable support–coverage trade-off: stronger evidential support accompanies a narrower published claim set, while exact reconstruction and complete communication of uncertainty remain unresolved. Future

work should connect detector-generated alarm windows to this downstream pipeline, improve retrieval and qualified-claim retention, and evaluate evidence-linked reports with independent analysts across additional OT environments.

References

- [1] E. Wagner, L. Bader, K. Wolsing, and M. Serror, “Sherlock: A dataset for process-aware intrusion detection research on power grid networks,” in *Proceedings of the Fifteenth ACM Conference on Data and Application Security and Privacy (CODASPY ’25)*. Association for Computing Machinery, 2025, pp. 419–424.
- [2] M. ur Rehman and H. Bahşi, “Process-aware security monitoring in industrial control systems: A systematic review and future directions,” *International Journal of Critical Infrastructure Protection*, vol. 47, p. 100719, 2024.
- [3] E. Hotellier, F. Sicard, J. Francq, and S. Mocanu, “Standard specification-based intrusion detection for hierarchical industrial control systems,” *Information Sciences*, vol. 659, p. 120102, 2024.
- [4] O. Koucham, S. Mocanu, G. Hiet, J.-M. Thiriet, and F. Majczyk, “Cross-domain alert correlation methodology for industrial control systems,” *Computers & Security*, vol. 118, p. 102723, 2022.
- [5] R. Smith, H. Janicke, Y. He, F. Ferra, and A. Albakri, “The agile incident response for industrial control systems (AIR4ICS) framework,” *Computers & Security*, vol. 109, p. 102398, 2021.
- [6] A. Staves, T. Anderson, H. Balderstone, B. Green, A. Gouglidis, and D. Hutchison, “A cyber incident response and recovery framework to support operators of industrial control systems,” *International Journal of Critical Infrastructure Protection*, vol. 37, p. 100505, 2022.
- [7] O. Arreche, T. R. Guntur, J. W. Roberts, and M. Abdallah, “E-XAI: Evaluating black-box explainable AI frameworks for network intrusion detection,” *IEEE Access*, vol. 12, pp. 23 954–23 988, 2024.
- [8] H. Xu, S. Wang, N. Li, K. Wang, Y. Zhao, K. Chen, T. Yu, Y. Liu, and H. Wang, “Large language models for cyber security: A systematic literature review,” *arXiv preprint arXiv:2405.04760*, 2024.
- [9] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, “Survey of hallucination in natural language generation,” *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.
- [10] E. Wagner, L. Bader, K. Wolsing, and M. Serror, “Sherlock: A dataset for process-aware intrusion detection research on power grid networks,” 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.18467070>